Methodology

This study was a descriptive cross-sectional analysis of publicly available UCMR 5 drinking-water monitoring data combined with development and implementation of a compound-specific clinical translation framework.

Data Source and Processing

Raw monitoring records from the EPA's Fifth Unregulated Contaminant Monitoring Rule (UCMR 5, 2023–2025) were obtained from publicly available EPA data files. UCMR 5 mandated PFAS testing across community water systems serving 3,300 or more people and a representative sample of smaller systems, collectively covering the large majority of the U.S. population served by public drinking water systems.(7) Records were linked to U.S. ZIP codes via Public Water System Identifier (PWSID) using a PWSID-to-ZIP crosswalk derived from EPA service area data. Non-detections were excluded from analysis. Where a water system served multiple ZIP codes, the same monitoring record was assigned to each served ZIP code; this reflects shared service-area geography and does not produce independent water-quality observations. The highest observed concentration per compound per ZIP code was retained across all monitoring periods; this approach was chosen to characterize the upper bound of observed drinking-water concentrations within each ZIP code rather than average long-term concentrations, and results should be interpreted accordingly. The resulting analytic dataset comprised 40,311 compound-level records spanning 12 analytes across 14,071 ZIP codes served by 4,885 unique water systems in all 50 states and U.S. territories. The ZIP code was the primary unit of analysis. ZIP codes in which all compounds returned below the minimum reporting level are not represented. Quality-control procedures verified the absence of duplicate compound entries per ZIP code and zero-value concentrations before analysis. Data processing and descriptive analyses were performed in Python (version 3.14.2). All processing steps are documented in a reproducible analysis script provided as supplementary material.

Statistical Analysis

Analyses were descriptive. Summary statistics included detection prevalence, exceedance prevalence, mean observed concentration, and maximum observed concentration per compound. No inferential statistical tests were performed, as the objective was to characterize observed monitoring results within the UCMR 5 dataset rather than estimate population parameters or test hypotheses.

Exceedance Characterization

For each compound with an established EPA MCL or Hazard Index threshold under the April 2024 National Primary Drinking Water Regulation (NPDWR), descriptive statistics were calculated as described above. The six regulated analytes and their applicable thresholds were: PFOA (individual MCL of 0.004 µg/L), PFOS (individual MCL of 0.004 µg/L), PFHxS (individual MCL of 0.010 µg/L), PFNA (individual MCL of 0.010 µg/L), HFPO-DA (individual MCL of 0.010 µg/L), and PFBS (regulated jointly with PFHxS, PFNA, and HFPO-DA via a combined Hazard Index with a threshold of 1, in which each compound's concentration is divided by its individual reference level and the four resulting fractions are summed).(5) Consistent with EPA regulatory practice, exceedance was defined as a detected concentration at or above the applicable MCL, not strictly above it. Because the dataset retains one record per compound per ZIP code rather than co-located sampling-event measurements, a full Hazard Index calculation requiring simultaneous multi-compound readings at the same sampling event was not feasible; PFBS is therefore reported by detection prevalence only.

PFOA and PFOS are reported as detection prevalence rather than exceedance rate. The UCMR 5 minimum reporting level for both compounds is 0.004 µg/L, which coincides with their respective MCLs by design; any recorded detection is therefore at or above the regulatory limit by definition, and an independent exceedance rate is not calculable from this dataset for these two compounds.(13)

For all remaining regulated compounds, the exceedance denominator was defined as the number of ZIP codes in the dataset with at least one detected concentration of that compound. Exceedance prevalence was calculated as the proportion of those ZIP codes with at least one concentration at or above the applicable MCL. This detection-conditional denominator is distinct from a population-weighted system-level denominator; the implications of this distinction are addressed in the Discussion.

Unregulated Analytes

Six of the 12 analytes in the dataset have no established federal MCL: lithium, perfluorobutanoic acid (PFBA), perfluoropentanoic acid (PFPeA), perfluoroheptanoic acid (PFHpA), perfluorodecanoic acid (PFDA), and 6:2 fluorotelomer sulfonic acid (6:2 FTS). Because no regulatory threshold exists for these compounds, all six are summarized by detection prevalence only. For lithium, detected concentrations are additionally reported relative to 7.36 µg/L, the exposure level examined in a prospective cohort study that reported an association with increased risk of autism spectrum disorder and schizophrenia in offspring,(12) and relative to the EPA's non-regulatory reference level of 10 µg/L.

Clinical Translation Framework

A compound-specific clinical translation framework was developed for all 12 analytes through structured review of peer-reviewed literature, National Academies reports, and EPA regulatory and toxicological documentation. For each compound, the framework documents the relevant toxicological mechanism and biological half-life where established; the applicable regulatory threshold or, for unregulated compounds, the most relevant evidence-based reference level; and published evidence on health effects at detected concentration ranges. Patient-facing context for each compound was calibrated to the detected concentration relative to the applicable threshold or reference level using this evidence hierarchy. Individual compound profiles were reviewed by a physician co-author with expertise in emergency medicine and epidemiology.

Web Tool Implementation

pfasestimator.org is a publicly accessible, no-registration web tool that accepts a U.S. ZIP code as input and returns the analytes detected in that ZIP code's water system, each compound's detected concentration relative to its applicable threshold, the corresponding compound profile from the clinical translation framework, and a plain-language summary generated by a large language model. Summaries are generated fresh for each user query using Llama 3.3 70B (Meta AI), conditioned on a structured, manually curated compound-specific evidence base containing toxicological mechanisms, regulatory thresholds, health effects, clinical context, and supporting references for each analyte. This evidence-conditioned architecture grounds responses in the curated evidence framework rather than permitting unrestricted free-form generation. Generation temperature and sampling parameters are determined by the backend implementation; outputs may vary across queries. The tool is freely accessible at pfasestimator.org.

Human Participant Protection

No protocol approval was necessary because this study did not involve human participants. All data used are publicly available, de-identified, system-level EPA monitoring records obtained from the UCMR 5 dataset.